



A systematic mapping study on automated negotiation for autonomous intelligent systems

Mashal Afzal Memon¹ · Gian Luca Scoccia² · Marco Autili¹

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Abstract

Autonomous intelligent systems are known as artificial intelligence software entities that can act on their own and can take decisions without any human intervention. The communication between such systems to reach an agreement for problem-solving is known as automated negotiation. This study aims to systematically identify and analyze the literature on automated negotiation from four distinct viewpoints: (1) the existing literature on negotiation with focus on automation, (2) the specific purpose and application domain of the studies published in the domain of automated negotiation, (3) the input, and techniques used to model the negotiation process, and (4) the limitations of the state of the art and future research directions. For this purpose, we performed a systematic mapping study (SMS) starting from 73,760 potentially relevant studies belonging to 24 conference proceedings and 22 journal issues. Through a precise selection procedure, we identified 50 primary studies, published from the year 2000 onward, which were analyzed by applying a classification framework. As a result, we provide: (a) a classification framework to analyze the automated negotiation literature according to several parameters (e.g., focus of the paper, inputs required to carry on the negotiation process, techniques applied, and type of agents involved in the negotiation), (b) an up-to-date map of the literature specifying the purpose and application domain of each study, (c) a list of techniques used to automate the negotiation process and the list of input to carry out the negotiation, and (d) a discussion about promising challenges and their consequences for future research. We also provide a replication package to help researchers replicate and verify our systematic mapping study. The results and findings will benefit researchers and practitioners in identifying the research gap and conducting further research to bring dedicated solutions for automated negotiation.

Keywords Autonomous intelligent systems · Multi-agent systems · Automated negotiation · Automated decision-making

Gian Luca Scoccia and Marco Autili have contributed equally to this work.

Extended author information available on the last page of the article

1 Introduction

The rapid evolution of artificial intelligence (AI) has given rise to autonomous intelligent systems that can perform actions without human intervention and make autonomous decisions (Zhang et al. 2017; Buiten 2019; Sharma et al. 2022; Zhu et al. 2024). Such systems not only function autonomously, but also communicate on behalf of their users in various domains, e.g., e-commerce, healthcare, autonomous vehicles, etc. (Levinson et al. 2011; Bagga et al. 2021; Kell et al. 2022).

When interacting with each other, these systems can be categorized into two broad categories: cooperative autonomous systems and competitive autonomous systems. Cooperative autonomous systems communicate and coordinate with each other to perform a shared task (Chen and Weiss 2019; Rothfuß et al. 2020); whereas competitive autonomous systems are selfish and only interact to maximize their own utility (Tsumura 2021; Gao et al. 2022). Therefore, how these systems interact with each other and autonomously resolve conflicts or, more generally, agree on matters of mutual interest is the focus of interest. One of the possibilities to resolve conflicts between autonomous systems is negotiation between these systems, which has gathered the research community's interest in automated negotiation, particularly within the domain of e-commerce (Sect. 4.1).

In general, negotiation is a process between multiple systems in which a decision is made jointly by communication, i.e., through the exchange of dialogues, bids, and offers to reach an agreement that is further accepted by all parties involved in the negotiation (Zuckerman et al. 2013; Bagga et al. 2022). Automated negotiation is a compelling field of research that groups three familiar research fields into one, namely, game theory, economics, and artificial intelligence (Baarslag et al. 2016). The significance of automated negotiation is receiving great attention in the present age, as intelligent agents who negotiate with each other and represent human users are likely to be more effective (An et al. 2016). Hereafter, the terms “system” and “agent” are used interchangeably throughout the text as an agent is a type of system that acts autonomously and makes decisions on behalf of users (Wooldridge and Jennings 1995).

To meet the growing demand for automated systems involving intelligent agents, the literature reports several reviews/surveys/mapping studies/secondary stuff focused on automated negotiation. Specifically, the survey by Kiruthika et al. (2020) groups the studies in the literature based on stages of the life cycle of a negotiation agent by considering opponent modeling only. The study presents a list of techniques used in the literature to model opponents during negotiation. As detailed below in Sect. 2, similar reviews of the negotiation literature have been previously proposed also by Lopes et al. (2008), Kraus (2001), Chen and Pu (2004), Khemakhem et al. (2022), Braun et al. (2006), Baarslag et al. (2016), Kell et al. (2022), Eshragh et al. (2015), Li et al. (2003)). However, until now, despite growing interest, the literature has not provided a comprehensive review of studies that *propose practical implementation of approaches or discuss techniques that can explicitly be employed to automate the negotiation process*. In addition,

the literature falls short in clearly reporting the main purpose and the application domain of the negotiation, which is needed to promote reuse and adoption. In this paper, we provide the research community with a comprehensive overview of the recent literature on automated negotiation, focusing on approach-oriented studies to offer practical insights into how negotiation functions within autonomous systems. That is, by narrowing our scope to these studies, we aim to highlight advances directly relevant to automation, giving a focused perspective that complements broader theoretical explorations in the field and identifying gaps where further innovation is needed. Although the selection process led to a relatively small number of studies, by focusing solely on these, we aim to provide the reader with findings that can directly contribute to the development and application of autonomous intelligent systems.

The goal of this systematic mapping study is to differentiate existing research in the field of automated negotiation from four distinct viewpoints: (1) the existing literature on negotiation with focus on automation, (2) the specific purpose and application domain of the studies published in the domain of automated negotiation, (3) the input, and techniques used to model the negotiation process, and (4) the limitations of the state of the art and future research directions.

To attain this goal, the methodology for systematic mapping study from Petersen et al. (2015), Wohlin et al. (2012) is applied, which concentrates on answering research questions about the state of the art by proposing a reproducible, objective, and unbiased approach. Summing up, the major contributions of our study are:

1. a classification framework for identifying and evaluating techniques and approaches for automated negotiation in the literature according to several parameters, e.g., focus of the paper, inputs required to carry on the negotiation process, techniques applied, and type of agents involved in the negotiation—*viewpoints (1)–(4)*;
2. an up-to-date map of the state of the art proposed in the domain of automated negotiation, identifying the main purpose, and application domain of each study—*viewpoint (2)*;
3. a list of techniques used for automating the decision-making process involving autonomous agents in a negotiation process, and a list of inputs required to carry out the negotiation process—*viewpoint (3)*;
4. a discussion about promising challenges and their consequences for future research on automated negotiation—*viewpoint (4)*;
5. a replication package for duplication and verification of our proposed study—*viewpoints (1)–(4)*.

The rest of the paper is organized as follows. Section 2 provides background knowledge of the negotiation process and lays out our study with related work. Section 3 presents the design of our study from a methodological point of view. The results of our study are presented in Sects. 4 and 5. Section 6 discusses future research challenges and threats to validity are discussed in Sect. 7. Section 8 concludes the mapping study.

2 Background and related work

In this section, we briefly discuss the negotiating agent's life cycle and discuss related studies in the literature. We focus on literature reviews, reports, and surveys that provide an overview of techniques that can be applied to automate the negotiation process. Table 1 presents the summary of the existing literature, comparing the findings of these studies with our work.

2.1 Background

Automated negotiating agents operate through a life cycle comprising three consecutive phases (Kiruthika et al. 2020; Braun et al. 2006), namely, *pre-negotiation*, *negotiation*, and optional *post-negotiation*.

During pre-negotiation, the agents focus on finalizing their attributes and personal preferences before initiating the negotiation. For example, in a negotiation scenario where the seller and buyer agents interact to carry out a purchase, the seller defines the value for preferred price that it would like to obtain and the value for the minimum reserved price below which it cannot accept any offer. Similarly, the buyer defines its values for the preferred price and the maximum reserved price. Moreover, the time limit to end the negotiation with no deal is also defined during this phase.

The negotiation phase comprises offer generation, opponent modeling, offer evaluation, and an acceptance model for decision-making (Kiruthika et al. 2020). During this phase, agents follow the negotiation protocol, which is a set of rules that defines how they interact with opponent agents. Few examples include the one-shot protocol in which one agent makes an offer that the opponent agent can either accept or reject (Ji et al. 2014), the alternative offer protocol where every agent can make an offer alternatively (Aydoğan et al. 2017), and multi-lateral negotiation protocol, in which participating agents take a decision cooperatively by agreeing mutually on a decision (Caillere et al. 2016).

The post-negotiation phase is optional. If the agents involved in the negotiation have found an agreement, its effectiveness can be evaluated during this phase in terms of agent utility or reward. For instance, in the same seller and buyer scenario mentioned above, the effectiveness of the decision can be evaluated by comparing the final offered value with the preferred value of both agents. If the final offered value is near the preferred value of both agents, it can be referred an effective decision (Bagga et al. 2022).

Agents can further be divided into *cooperative*, when negotiating to find a mutually beneficial solution, and *non-cooperative*, when acting independently and maximizing their own utility. Furthermore, agents can have complete or incomplete information about the negotiation setting, e.g., other agent's preferences (Li et al. 2003).

Table 1 Characteristics of the related work and details about primary studies considered

Related work	Characteristics of the related work			Details about primary studies considered in the related work						
	Type	Time frame	Main focus	Uses systematic approach	Provides classification framework	Covers studies explicitly focused on automation	Differentiates negotiation purpose of covered studies	Specifies application domain	Reports on negotiation inputs	Identifies limitations
(Lopes et al. 2008) Negotiation among autonomous computational agents: principles, analysis and challenges. Artificial Intelligence Review	Analytical Study	Until 2008	Breaking the negotiation process into phases like pre-negotiation, actual negotiation, and renegotiation	✗	✓	✓	✗	Scattered, not a consolidated list	Only for few studies. Not a comprehensive list	Scattered in text, not a consolidated list
(Li et al. 2003) A review of research literature on bilateral negotiations. Technical report, Robotics Institute, Carnegie Mellon University	Review	Until 2003	Covering literature on bilateral negotiation	✗	✗	✗	✗	Navy	Scattered for few studies	Scattered in text, not a consolidated list
(Kraus 2001) Automated negotiation and decision making in multiagent environments. ECCAI advanced course on artificial intelligence	Survey	Until 2001	Focuses on negotiation of self-interested agents, with a little focus on cooperative agents	✗	✗	✗	✗	Economics	✗	✗

Table 1 (continued)

Related work	Characteristics of the related work			Details about primary studies considered in the related work						
	Type	Time frame	Main focus	Uses systematic approach	Provides classification framework	Covers studies explicitly focused on automation	Differentiates negotiation purpose of covered studies	Specifies application domain	Reports on negotiation inputs	Identifies limitations
(Chen and Pu 2004) Survey of preference elicitation methods. Technical Report, Swiss Federal Institute of Technology	Survey	Until 2004	Covering literature on preference elicitation techniques	✗	No, but organizes data thematically	✗	✗	E-commerce	✓	✓
(Khemakhem et al. 2022) Agent-based intelligent decision support systems: systematic review. IEEE Transactions on Cognitive and Developmental Systems	Systematic Review	Until 2019	Reviews 58 studies focused on intelligent decision support systems	✓	✓	✗	✗	Medicine, supply chains, e-commerce	✗	✗
(Braun et al. 2006) E-negotiation systems and software agents: Methods, models, and applications. Intelligent Decision-making Support Systems	Review	Until 2004	Understanding the methods, models, and applications of e-negotiation systems and software agents	✗	✗	✗	✗	Planning, e-commerce, scheduling	✗	Scattered in text, not a consolidated list

Table 1 (continued)

Related work	Characteristics of the related work			Details about primary studies considered in the related work						
	Type	Time frame	Main focus	Uses systematic approach	Provides classification framework	Covers studies explicitly focused on automation	Differentiates negotiation posture of covered studies	Specifies application domain	Reports on negotiation inputs	Identifies limitations
(Kiruthika et al. 2020) Lifecycle model of a negotiation agent: A survey of automated negotiation techniques, Group Decision and Negotiation	Survey	Until 2018	Covers literature based on the life cycle of negotiation agents and provides taxonomy of opponent modeling techniques	✗	No, but organizes data based on negotiating agent's life cycle	✓	✗	✗	✗	Scattered in text, not a consolidated list
(Baarslag et al. 2016) Learning about the opponent in automated bilateral negotiation: a comprehensive survey of opponent modeling techniques, Autonomous Agents and Multi-Agent Systems	Survey	Until 2014	Provides a review of literature on opponent modeling	✗	✓	✓	✗	✗	✗	✗
(Kell et al. 2022) Machine learning applications for electricity market agent based models: A systematic literature review, Pre-print	Systematic literature review	Until 2021	Focuses on machine learning techniques applied to agent-based electricity market models	✓	✗	✓	✗	Economics	✗	✓

Table 1 (continued)

Related work	Characteristics of the related work				Details about primary studies considered in the related work					
	Type	Time frame	Main focus	Uses systematic approach	Provides classification framework	Covers studies explicitly focused on automation	Differentiates negotiation purpose of covered studies	Specifies application domain	Reports on negotiation inputs	Identifies limitations
(Eshragh et al. 2015) Automated negotiation in environmental resource management: Review and assessment. Journal of Environmental Management	Review	Until 2015	Describes main components of automated negotiation, reviews and compares machine learning techniques	✓	✓	✓	✓	Environmental resource management	✓	✓
(Zhu et al. 2024) Advancing autonomy through lifelong learning: a survey of autonomous intelligent systems. Frontiers in Neurobotics	Survey	Until 2024	Focuses on lifelong learning techniques for intelligent autonomous systems	✓	✓	✓	✓	✓	✓	✓
(Grislin-Le Strugeon et al. 2022) A systematic mapping study on agent mining. Journal of Experimental & Theoretical Artificial Intelligence	Systematic mapping study	Until 2020	Focuses on capabilities of software agents for data mining	✓	✓	✓	✓	E-commerce, marketing, medicine, social sciences, E-learning	✓	Scattered in text, not a consolidated list

Table 1 (continued)

Related work	Characteristics of the related work			Details about primary studies considered in the related work						
	Type	Time frame	Main focus	Uses systematic approach	Provides classification framework	Covers studies explicitly focused on automation	Differentiates negotiation purpose of covered studies	Specifies application domain	Reports on negotiation inputs	Identifies limitations
Our work	Systematic mapping study	Until 2022	Providing the research community with a comprehensive overview of the recent literature on automated negotiation, particularly emphasizing studies that propose approaches for automating the negotiation process	✓	✓	✓	✓	✓	✓	✓

2.2 Related work

In the following, we present the studies related to our work, positioning our work within the context, to highlight our contributions. The findings are summarized in Table 1.

The proposed study by Lopes et al. (2008) highlights the framework of automated negotiation that encompasses various stages, including pre-negotiation, negotiation, and re-negotiation, and focuses on techniques used to select negotiation strategies and choose a negotiation protocol to carry out the process. As presented in Table 1, similar to our work, the study presents a classification framework that categorizes the literature as theoretical or computational studies. Moreover, it provides a discussion of the underlying concepts and notion employed in the negotiation process such as *bilateral or multilateral negotiation*, *single issue or multi-issue negotiation*. The study explains attributes of the negotiation process; however, it is state-of-the-art as it mostly covers studies from the last decade. Compared to this study, our study covers the recent literature which focuses on automating the negotiation process using a systematic approach. Our work offers a comprehensive overview of the negotiation literature by adopting the classification framework outlined in Table 3, which organizes the literature according to various categories, including the purpose of the negotiation, the application domain, the techniques and inputs required to carry out the negotiation and the limitation of each study.

Bilateral negotiation is a scenario in which only two agents are involved in the negotiation. The literature review by Li et al. (2003) provides a detailed understanding of bilateral negotiation, considering it as a game theory problem. The study discusses several scenarios of bargaining for price and provides a better understanding of bilateral negotiation in economics and AI. Still referring to Table 1, the study underscores its application domain in the Navy detail system, which operates as a matching market where sailors and commands engage in negotiations similar to bargaining scenarios observed in price-based negotiations. Consequently, the literature focuses on the literature centered on economics. In contrast, our mapping study covers the literature that not only focuses on economics but also covers other application domains, detailed in Sect. 4.1. In addition, the study lacks the review of the recent state of the art and is limited only to the techniques used for bilateral negotiation. Our study covers the recent literature with a more general focus on automated negotiation involving the literature for bilateral as well as multilateral negotiation.

Complementing the previous study, the survey by Kraus (2001) focuses on approaches to negotiation in multi-agent environments where agents are considered self-interested. It also mentions the techniques for cooperative agents. The proposed study considers negotiation time, efficiency, and simplicity of negotiation strategies as attributes for evaluating decisions. For example, negotiation time measures the delay in reaching an agreement, which causes more computational resources compared to the ones that reach an agreement early. Similarly, the efficiency of the decision can be evaluated by comparing the reward of all agents involved in the negotiation, which should be equal for every agent to reach a successive agreement, and simplicity can be evaluated by computing the time duration for generating strategies. The primary focus of the proposed study is on strategic negotiation for price-based

negotiations with in economics. In contrast, our systematic mapping study focuses on multiple goals to automate the negotiation process focusing on multiple application domains.

Preference elicitation is a prerequisite before starting the negotiation process. As mentioned in Sect. 2.1, the agents can finalize their preferences during the pre-negotiation phase. Besides that, preference elicitation is also useful when an autonomous agent negotiates on behalf of the human agent, since the goal of the automated decision system is to assist humans. Hence, it is necessary for autonomous agents to accurately model user preferences. A survey by Chen and Pu (2004) explains preference elicitation techniques, such as a value function that queries the user to rank the outcome of the value function and clustering that combines and compares the preferences of different users. Although the study provides some techniques for preference elicitation, it lacks the techniques presented in the recent literature that focus on the agent's history of offers to predict user preferences and represent them in automated negotiation. Our study not only covers the literature that focus on preference elicitation but also explores studies that address different purposes of negotiation. Moreover, as presented in Table 1, the study by Chen and Pu (2004) organizes the negotiation literature around specific parameters without explicitly mentioning the classification framework. However, these studies majorly focus on a single aspect of negotiation. In contrast, the classification framework in our study is designed to be more comprehensive and generic, addressing multiple parameters.

A recent study by Khemakhem et al. (2022) provides a systematic review of decision-making systems based on multiple distributed agents. These systems employ a conflict resolution strategy to overcome those cases in which discording answers are given by the different agents that compose the system. Similar to our work, as presented in Table 1, the study follows a systematic approach and presents a classification framework aimed at analyzing the application of agents. However, unlike our work, the study focuses on conflict resolution strategies in general and reports that negotiation is one of the techniques to resolve conflicts. The adopted negotiation techniques and their characteristics are not reported. Our work, introduces a list of techniques employed in the recent literature to automate the negotiation process.

The study by Braun et al. (2006) provides a review of software agents for electronic negotiation. This study provides a detailed review of the types of agents, the tasks these agents can perform, the techniques that negotiator agents can employ during the negotiation process, and the applications of these agents e.g., planning, e-commerce, and scheduling, as presented in Table 1. However, similar to most studies in the literature, the study by Braun et al. (2006) also covers the negotiation literature of the last decade. Unlike these, our work adopts a systematic approach to identify and analyze the most relevant literature in the field that focuses on automating the negotiation process.

Similarly, the authors (Kiruthika et al. 2020) propose a new model for the life cycle of a negotiation agent, categorize the existing negotiation literature based on the agent's life cycle, and provides a list of techniques that agents use for modeling the opponent to predict their preferences during negotiation. As reported in Table 1, the study also organizes the negotiation literature based on the agent's life cycle. No classification framework is adopted. The study focuses primarily on techniques used

to model the opponent; our study covers a broader range of techniques with multiple purposes, including opponent modeling, offer generation, and other key aspects.

The results of a closely related survey are reported in Baarslag et al. (2016), focusing on techniques that can be used for opponent modeling. The authors provide an in-deep discussion on how opponent models benefit negotiation agents. The study also presents a classification framework that extracts insights from studies, focusing only on the parameters by which opponents are learned. However, similar to the study by Kiruthika et al. (2020), this study also focuses on a single aspect of negotiation. Differently from these studies, our work relies on a systematic process for the discovery of recent relevant literature and explores techniques employed by agents to automate the entire negotiation process.

A recent study by Kell et al. (2022) follows a systematic approach to provide a review of machine learning techniques that can be used for bidding strategies in the electricity supply market. Similarly, the study by Eshragh et al. (2015) provides a review of machine learning techniques for automated negotiation in the environmental resource management domain. In relation to the two studies above, ours is broader in scope, covering studies that apply negotiation to a variety of domains (see Sect. 4.1) and can possibly rely on negotiation techniques not grounded in machine learning (see Sect. 4.2).

A survey is presented in the study by Zhu et al. (2024), which offers a detailed review of autonomous intelligent systems that employ lifelong learning. This approach enhances the system adaptability, enabling it to continuously learn and improve their performance in various environments. Although, similarly to us, the survey focuses on autonomous intelligent systems that operate independently, this work focuses solely on lifelong learning techniques for their functionality with no consideration of negotiation. In contrast, our work focuses on multiple techniques these systems can employ for negotiation with other systems in various application domains.

Closely to our systematic approach, the study by Grislin-Le Strugeon et al. (2022) investigates the potential of software agents for data mining. The study also presents a classification framework to extract the data from studies through certain parameters. While the design of the study aligns closely with our approach, the investigation conducted by Grislin-Le Strugeon et al. (2022) concentrates on autonomous agents that perform data mining, whereas our research emphasizes automated negotiation between intelligent agents.

For what specifically concerns the usage of a classification framework, in the following we summarize its adoption in existing literature. The study by Baarslag et al. (2016) introduces a classification framework to extract insights from research, emphasizing the parameters used to learn about opponents only. Similarly, the work of Lopes et al. (2008) presents a framework that categorizes studies as either theoretical or computational, while Khemakhem et al. (2022) proposes a classification framework for analyzing the application of agents. The studies by Kiruthika et al. (2020), Chen and Pu (2004) structure negotiation literature around specific parameters, albeit without explicitly referencing a classification framework. All these works predominantly focus on individual aspects of negotiation. In contrast, our classification framework is designed to be more comprehensive and versatile, addressing

multiple parameters, such as the techniques employed, the inputs required for negotiation, its purpose, and application domain. As highlighted in Table 1, similar to our approach, the studies by Lopes et al. (2008), Kiruthika et al. (2020), Baarslag et al. (2016), Kell et al. (2022), Eshragh et al. (2015) also emphasize the role of automation in negotiation research. However, they are limited to addressing specific purposes of negotiation, which distinguishes our contribution.

Summing up, in this section, we mentioned several surveys, literature reviews and reports. Although some of the studies are performed systematically and introduce classification frameworks, similar to our study. These studies mainly consider the literature from the last decade. Moreover, similar to our work (Table 1), the studies by Lopes et al. (2008), Kiruthika et al. (2020), Baarslag et al. (2016), Kell et al. (2022), Eshragh et al. (2015) emphasize the automation of negotiation within the literature; however, they are limited to addressing specific purposes of negotiation, which distinguishes our contribution. In addition, our study distinguishes whether the negotiation is carried out between humans and agents or it is entirely automated, and offers insights into the discussion on how such negotiations could become fully autonomous. Our work also presents the limitations of each study to provide the reader with better insight and direction for future investigations (see Sect. 6). In the next section, we describe the process we carried out for our systematic mapping study.

3 Study design

For our systematic mapping study, we structured our research into three main phases, commonly employed in systematic literature studies (Kitchenham and Brereton 2013; Wohlin et al. 2012; Grislin-Le Strugeon et al. 2022): planning, conducting, and documenting.

Planning—During this phase, we realized (1) the need to perform a systematic mapping study on automated negotiation for agreement, (described in Sect. 2.2), and (2) we defined the research questions (further discussed in Sect. 3.1).

Conducting—At this stage, we followed the required steps to perform this mapping study such as (1) searching for and selecting relevant studies for automated negotiation (Sect. 3.2), (2) scrutinizing relevant studies and extracting data from each one following an analysis framework (Sect. 3.3), and (3) synthesizing the main insights emerging from the data collected during the previous activity (Sect. 3.4).

Documenting—In the last stage, we finalized the process by (1) an in-depth elaboration of the data extracted in the previous phase, with the main goal of setting the obtained results in their context, (2) having a detailed discussion of possible threats to validity, and (3) writing the final report (i.e., this article).

A complete replication package is publicly available to allow interested researchers to independently replicate and verify our study.¹ The package includes a comprehensive data extraction form, the raw extracted data, and the scripts employed for data analysis.

¹ <https://github.com/mashalafzal/Automated-negotiation-replication-package>.

3.1 Research questions

The aim of this study is to help the research community to summarize the state of the art on automated negotiation that proposes practical implementation approaches, while highlighting existing limitations, open challenges, and future research directions. Specifically, the goal of our mapping study is formulated by using the Goal-Question-Metric perspectives (i.e., purpose, issue, object, viewpoint (Caldiera and Rombach 1994)). Table 2 shows the resulting outcome. We further divided our goal in the following research questions:

RQ1 What is the specific purpose of the negotiation?

With this research question, we aim to identify and categorize the reasons for which negotiation is employed in the literature, to provide comprehensive list of the negotiation purposes, and its possible application domains.

RQ2 Which inputs (RQ2.a) and techniques (RQ2.b) are employed to carry out the negotiation?

This research question explores how the negotiation process is carried out. Specifically, we investigate (a) what inputs are needed to carry out the negotiation, and (b) what techniques are applied to automate the negotiation process.

RQ3 What are the limitations of the current approaches, open issues, and future challenges?

Automated negotiation is a promising research topic. However, to reap its benefits, there are still limitations and deficiencies that need to be overcome to enable its practical application in a real-world scenarios. Hence, this research question focuses on assessing and categorizing the limitations and open issues present in the current state of the art, and addresses the future work to overcome these limitations.

3.2 Search and selection process

An overview of our search and selection process is shown in Fig. 1. The process is responsible for the inclusion or exclusion of any potential study for this research. It is a multi-stage process whose steps are described below.

Manual keyword search— We have started our search and selection procedure by identifying the top-level venues for computer science, software engineering and artificial intelligence research. By focusing on interdisciplinary venues, we ensure a selection of studies that interest in areas such as automation and intelligent systems, aligning with the aim of our study. For the database selection, we selected

those well-known for their rigorous selection criteria, which ensure that the studies included are from well-regarded venues. Specifically, for conferences, we adopted the *GII-GRIN-SCIE-Conference-Rating*² ranking as a reference list and selected from it all conferences ranked A or higher in the domains of interest. *GII-GRIN-SCIE (GGS)* is a conference ranking list initiated by a group of *Italian and Spanish* Professors in the domain of *computer engineering and computer science*. It provides the bibliometric approach for the conference proceedings that are not available in *SCOPUS* database. For journals, we used the *SCOPUS* database³ as a reference and selected from it all potentially relevant journals by searching from the top 10% journals in subject areas of interest for our research (i.e., *computer science, artificial intelligence, software engineering, computer science applications, and general computer science*). This selection of top-level venues resulted in the identification of 24 conferences and 22 journals in which to search in-depth for studies potentially relevant to our research. The full list of selected venues is available in Table B.

Inside these venues, we performed a keyword-based search within the paper titles, using the following search query: (*negot* OR *agree*). These keywords have been selected as they naturally represent the root form of words frequently found in titles of studies in the field of automated negotiation (i.e., *negot* for negotiate and negotiation, *agree* for agreement). By using the wide-coverage search query above, we can reasonably suppose that we captured all the relevant papers, while ensuring their close relation to the focus of our study. A similar approach was adopted in Khemakhem et al. (2022), employing a set of keywords to capture studies relevant to multi-agent systems. The search was carried out inside the Computer Science Bibliography from the University of Trier (i.e., DBLP (Ley 2002)), considering a time frame that goes from 2017 to 2022 for each of the venues selected at the previous step. This search considered a total corpus of 73,760 studies, out of which 56,819 were the results of conference proceedings and 16,941 belong to journal publications. The selected keywords were found in the titles of 140 potentially relevant studies, where 118 studies have been selected as a result of searching in conference proceedings and 22 studies have been selected as a result of searching journals.

Studies selected during this keyword-based search were further classified and sorted according to the parameters of our inclusion and exclusion criterion discussed below. It is worth mentioning that this step was not only focused on extracting *all* relevant studies but, as suggested in Wohlin (2014), we concentrated on extracting relevant studies to utilize as a basis for a subsequent snowballing procedure, discussed in the following.

Apply selection criteria— The resulting studies were further filtered out according to inclusion and exclusion criteria discussed in Sect. 3.2.1. Following the guidelines in Petersen et al. (2008a), Zhang and Babar (2013), we carried out the selection process in a time-efficient and objective manner, employing an adaptive reading depth procedure as it was not necessary to read the full text of studies that did not qualify. For instance, the study by Cardenas and Kim (2020) appeared as a result of

² <https://scie.lcc.uma.es:8443/>

³ <https://www.scopus.com/sources>.

Table 2 Goal of this research

Purpose	Identifying and accessing
Issue	the techniques and approaches used for automated negotiation
Object	in the state of the art for automated decision making
View Point	from a researcher's and practitioner's point of view

the search query, as one of the keywords *agree* is found in the title, but was excluded due to its irrelevance to the main focus of the study. Similarly, studies that did not fulfill the inclusion and exclusion criteria were eliminated, resulting in an initial selection of 22 studies. These studies were used as a basis for subsequent snowballing to find additional relevant studies in the next step. This reduction in the number of results is expected, as the selected venues are broad in scope, covering varying topics on automation and intelligent systems. We acknowledge that the selection criteria resulted in a reduction of the number of studies selected. However, as mentioned above in Sect. 1, this ensures that our focus remains on directly relevant studies concerning automation, thereby offering the community a thorough review that supports advances in the field.

Backward and forward snowballing— To have a comprehensive set of studies representative of the literature of the field, we applied *backward and forward snowballing* (Wohlin 2014). For snowballing, we also applied the keyword-based search using the *negot* and *agree* keywords followed by the selection criteria. We used *backward snowballing* to evaluate the studies that are cited by the set of our initially selected studies. Backward snowballing allowed us to identify studies related to our study's aim that were not published in the selected venues. This stage resulted in a total of 766 potentially relevant studies. The high number of 766 studies gathered from backward snowballing includes all studies references by the initially selected 22 studies, spanning many years, including those published before the time frame of our initial search. In addition, many of these studies were broad in scope and did not align directly with the focus of our study, with several studies commonly cited by the initially chosen studies. For instance, the study by Mnih et al. (2015) appeared as a result of backward snowballing PS19. Although PS19 cites this research to explain reinforcement learning, it lacks a discussion on the automation of negotiation, which is the main focus of our study. Subsequently, we applied the keyword-based search and inclusion and exclusion criterion to these results, and 19 relevant studies have been selected as a result of backward snowballing. As mentioned above, although the backward snowballing yielded a large number of studies, our emphasis on automation within the selection criteria led to a limited number of selected studies, aligning with the main objective of the study.

Furthermore, *forward snowballing* (Wohlin 2014) has also been applied by analyzing the studies that have cited the set of our initially selected studies. Through this step, we figure out all those new and relevant studies that were not included in conference proceedings or journal issues. Forward snowballing yielded the total number of 279 potentially relevant studies, which were also further scrutinized according to

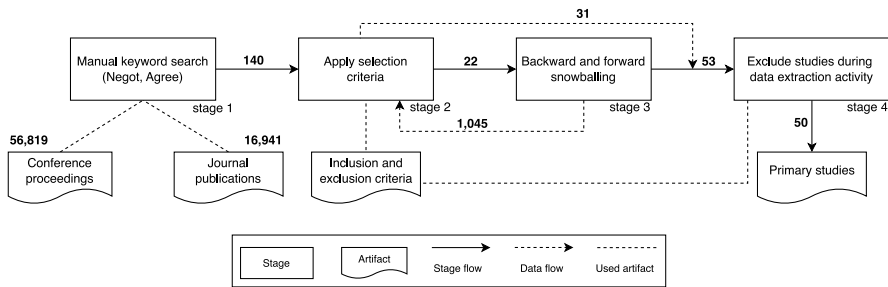


Fig. 1 The search and selection process of our study

the keyword-based search and inclusion and exclusion criterion mentioned below in Sect. 3.2.1. Duplicates that appeared during this step were filtered prior to applying the inclusion and exclusion criteria. Consequently, after applying the selection criteria, 22 studies have been selected as the result of forward snowballing.

To carry out both backward and forward snowballing, we have utilized the *Google Scholar*⁴ database. Following the inclusion and exclusion criterion, we have figured out that 41 out of 1,045 results overall snowballing were actually relevant for our study. Initially, the snowballing results were not cross-checked against the selected 21 studies to ensure consistency in the data extraction and to confirm that our approach remained thorough. However, when cross-checked, out of 41, 10 studies were already included in the set of our initially selected studies. Therefore, a total of 31 studies were chosen through the snowballing process, in addition to 22 primary studies scrutinised as per our inclusion and exclusion criteria based on the initial search outcomes, resulting in a sum of 53 studies.

Exclude studies during data extraction activity—During the data extraction, we also paid attention to the duplicate studies present in the set of selected studies, i.e., papers that were found to be the same but published in multiple venues. For instance, Bagga et al. (2021) and Bagga et al. (2020) focus on opponent modeling for counter offer generation using deep reinforcement learning. These studies were merged as PS49 (a) and PS49 (b) as both studies propose the same technique, published by the same authors in different venues. A total of 3 pairs of studies were merged this way, resulting in a final set of 50 primary studies selected for further analyses. The merged studies are counted as one throughout the paper, except for Sect. 5, where the studies are counted separately according to the inter parameter mapping. The final list of selected primary studies is available in Table A.

3.2.1 Selection criteria

In order to minimize the possibility of biases, we followed well-known guidelines to conduct systematic literature reviews (Kitchenham and Brereton 2013; Petersen et al.

⁴ <https://scholar.google.com/>

2008a, 2015) and defined *inclusion* and *exclusion criteria* prior to performing the selection of relevant studies. The selection criteria is aligned with the study's aim, focusing on studies that present the implementation of automation approaches, thus ensuring only relevant studies are selected. In the following, we detail the set of inclusion and exclusion criteria that guided the selection. Notice that, any study which satisfied *all* inclusion criteria was included. However, any of the studies which satisfied *at least one* of the exclusion criteria was discarded.

Inclusion criteria

- I1 Studies in which autonomous systems take decisions as a result of agreements reached through negotiation processes.
- I2 Studies employing techniques to automate the negotiation process (e.g., through machine learning).

Exclusion criteria

- E1 Studies that consider only human-to-human interaction without involving any autonomous system.
- E2 Studies not peer-reviewed, such as preprints and technical reports.
- E3 Secondary studies, such as surveys and literature reviews.
- E4 Studies lacking a technical description of the proposed approach (e.g., fully theoretical or high-level description only).
- E5 Studies not written in English.

3.3 Data extraction

During this stage: (1) the classification framework was designed for the data extraction activity, and (2) the primary studies selected during the previous steps were further scrutinized to extract meaningful data.

To perform the data extraction process, a predefined data extraction form was designed prior to beginning the data extraction process. The data extraction form structure reflects the various categories of the classification framework. The classification framework is composed of three distinct parts, one for each research question of our study. The overview of the classification framework and the respective parameters is reported in Table 3.

Additionally, we read the sections related to the case study (detailing application scenarios), the experimental setup (providing technical details), and the discussion to extract the application domain of each study.

An adaptive reading depth was applied in order to carry out the data extraction process in a time-efficient and objective manner (Petersen et al. 2008b): we began by reviewing the abstract, introduction and motivation, and conclusion sections of the selected primary studies to determine the specific purpose of the negotiation and the application domain. If needed, we further read the sections related to the case study, experimental setup, or discussion, when available. Later, we classified the studies into groups based on their purpose of negotiation and application domain.

Table 3 Overview of the classification framework

RQ	Goal	Parameters
RQ1	Specific Purpose	Negotiation purpose Application domain
RQ2	Input, Techniques	Negotiation inputs Techniques used for automation Type of agent involved in negotiation
RQ3	Limitations, Future Study	Limitations and future research

The results of each parameter, alongside the discussion of the extracted data, are given in Sect. 4.1.

Subsequently, we extracted the techniques used to automate the negotiation process and the inputs needed by the agents to carry on the negotiation by analyzing sections of each paper that provide a technical description of the proposed approach, e.g., methodology, algorithm description, and experimental setup. The scrutinized results are presented in Sect. 4.2.

To assess the limitations of each study, we examined the sections such as discussion, conclusion and future work. Indeed, by convention, limitations and future work of a study are generally listed and discussed in these sections. In Sect. 4.3, we present the results of our data extraction activity detailing the limitations found among the selected primary studies and proposing possible solutions.

Data extraction was performed independently by two researchers, each performing an evaluation of the selected studies, followed by a collaborative analysis. The two researchers collaborated to construct a record for each analyzed study that summarizes the extracted information and allows further analysis.

3.4 Data synthesis

This stage deals with aggregating and summarizing the data extracted from the primary studies as suggested in Keele et al. (2007). Here, the main goal is understanding, analyzing, and characterizing the current state of the art of research for automated negotiation.

We further divided the process of data synthesis into two stages, i.e., vertical and horizontal analysis. For the *vertical analysis*, we examined the extracted data to look for possible trends and patterns for each parameter of our classification framework. Instead, for the *horizontal analysis*, we considered parameters in pairs, to discover possible relationships across parameters. For this purpose, we used contingency tables by applying the same process utilized in the study by Autili et al. (2021). In both phases, we performed a combination of content analysis (for categorizing and coding the studies under broad thematic categories) and narrative synthesis (mainly for explaining results in detail and interpreting the findings).

4 Results

This section provides a detailed explanation of parameters involved in the classification framework and illustrates the related data extracted from primary studies. As mentioned above in Sect. 3.2, each of the three duplicated studies, together with their merged counterparts, is considered a single study throughout the paper except in Sect. 5. Consequently, the results below are based on a total of 50 studies.

4.1 Specific purpose (RQ1)

In the following, we discuss the results for parameters related to the purpose for which negotiation is employed.

Negotiation purpose— The negotiation purpose represents the principal goal for which the negotiation techniques were conceived. By analyzing the primary studies as mentioned above in Sect. 3.3, four main purposes categories emerged from the keywording process: *reward maximization*, *opponent modeling*, *agreement for cooperation*, and *preference elicitation*. Table 4 summarizes the primary studies divided by their goal and, in the following, we discuss each of the four goals in detail. Some of the studies focus on more than one goal which is also given in Table 4.

Example: The primary study PS49, proposes an approach to learn opponent strategies using deep neural networks for autonomous negotiation in e-Commerce. Moreover, it generates counter offers based on the learned strategies of the opponent to secure more rewards compared with its preferred value. Similarly, other studies also focus on more than one goal and are represented in Table 4 under multiple categories of the purpose.

Reward maximization – The majority of the primary studies (40/50) propose techniques aimed at maximizing the reward tied to the offers generated by the agents. Particularly, most of the proposed studies focus on negotiation for a price,

Table 4 Studies grouped by purpose

Purpose	Studies #	Primary studies
Reward maximization	40	PS2, PS5, PS6, PS7, PS8, PS9, PS10, PS11, PS12, PS13, PS14, PS16, PS17, PS19, PS20, PS21, PS22, PS23, PS24, PS25, PS26, PS27, PS28, PS29, PS30, PS31, PS32, PS34, PS35, PS36, PS37, PS38, PS39, PS41, PS42, PS43, PS45, PS46, PS47, PS49
Opponent modeling	15	PS4, PS6, PS7, PS11, PS14, PS18, PS19, PS21, PS23, PS33, PS40, PS43, PS44, PS49, PS50
Agreement for cooperation	6	PS1, PS3, PS5, PS9, PS15, PS48
Preference elicitation	6	PS24, PS27, PS32, PS34, PS36, PS38

leveraging a utility function to maximize the reward for the agents, depending on opponent offers and the time elapsed since the beginning of the negotiation.

Example: The authors of PS8 argue that agents can increase their reward by taking advantage of the negotiation context, i.e., the set of facts, events, trends, and special circumstances that surrounds the negotiation. To this end, they propose a context-aware negotiation agent that learns through self-play and reinforcement learning how to use key contextual information to gain a competitive edge over other agents.

Opponent modeling – Information about the opponents is essential to improve automated negotiation strategies (Hindriks et al. 2009). Understanding the behaviour of opponents is referred to as opponent modeling and helps in finding a more successful outcome of the negotiation for both parties involved. A total of 15 primary studies propose techniques that aim to improve the current state of the art for opponent modeling.

Example: The primary study PS19 focuses on opponents' history for offer generation in order to learn how to maximize the chances of acceptance of an offer during negotiation. It proposes a novel technique for automated negotiation agents where an agent uses Q-learning to develop a bidding strategy until the agreement is reached between the two parties involved in the negotiation process.

Agreement for cooperation – While negotiation is often employed with the goal of maximizing utility, other studies focus on reaching an agreement between autonomous agents to perform a task cooperatively. We group the latter studies under this goal. A total of 6 primary studies fall in this category.

Example: The study PS5 introduces an approach called distributed emergent agreement learning (DEAL). It enables agents suggest agreements to other agents in order to complete a task more efficiently. The approach has been evaluated in the context of a smart factory, where agents perform a task with cooperation through bilateral agreement and divide the reward in exchange for services provided.

Preference elicitation – In the context of automated negotiation, designing agents that can effectively acquire and integrate users' preferences into their own decision-making is an open research direction. We group under this goal studies that propose novel preference elicitation methods, to better capture the users' preferences, in both an autonomous or assisted manner. A total of 6 primary studies fall into this goal.

Example: PS36 introduces a novel agent-based approach to negotiate the permission to access private data between users and online services. The agent proposed by the authors autonomously negotiates potential agreements for the user, which they can refine by manually continuing the negotiation. The agent learns from these interactions and updates the user model in subsequent interactions.

Application domain— This parameter is used to represent the field of application for the techniques proposed by the primary studies. As mentioned above in Sect. 3.3, the data extraction resulted in five distinct categories in which negotiation approaches are employed: *economics*, *generic/unspecified*, *cooperative environment*, *competing environment* and *privacy*. The distinction between cooperative and competitive domains is due to the fact that in a cooperative environment, agents engage in negotiation to reach a shared goal (Bachrach et al. 2020). In contrast, in a competitive environment, agents strive to attain their own objectives and prioritize their own interests, disregarding any shared goal (Mansour

Table 5 Studies grouped by application domain

Domain	Studies #	Primary studies
Economics	27	PS2, PS8, PS10, PS12, PS14, PS16, PS17, PS20, PS22, PS26, PS27, PS28, PS30, PS31, PS32, PS34, PS35, PS37, PS38, PS39, PS41, PS42, PS43, PS45, PS46, PS47, PS49
Generic/unspecified	14	PS6, PS11, PS13, PS18, PS19, PS21, PS23, PS24, PS25, PS29, PS33, PS40, PS44, PS50
Cooperative environment	6	PS1, PS3, PS5, PS9, PS15, PS48
Competing environment	3	PS4, PS7, PS16
Privacy	1	PS36

2020). In Table 5, a breakdown of the studies grouped by application domain is provided. A total of 27 studies focus on techniques to negotiate on the price of a product or a service. Hence, we summarize the application domain of these studies as *economics*. A total of 14 studies instead generate offers without considering a specific domain, as they propose generic techniques that can be used to negotiate in any environment, depending on the goal of the agent. Thus, these techniques have been labeled as having a *generic/unspecified* domain. Moreover, 6 primary studies propose techniques for a *cooperative environment*, where the agents negotiate in order to perform a task cooperatively and share the reward afterward. Conversely, 3 studies focus on a *competing environment*, in which the agents consider each other as opposing players and generate offers accordingly. Finally, a single study focuses on the *privacy* domain, in which the object of the negotiation is the access permissions granted to social network apps on behalf of the human user.

4.2 Input and techniques (RQ2)

In the following, we discuss the results related to the technical details of the inputs (RQ2.a) and techniques (RQ2.b) used in primary studies.

Negotiation inputs— This parameter is used to summarize the inputs required by the techniques proposed in the primary studies. With reference to Table 6, the majority of the studies (32) take as input *reserved, preferred and initial price values* required in order to negotiate on the price of products or services. Another commonly used input is the *opponents' offers history*, employed by 15 primary studies. This information allows to improve on the generation of offers by considering how opponents have acted in the past. Similarly, 6 studies take as input the history of offers of the user on behalf of which they are negotiating, in order to understand his behavior and preferences. Furthermore, 6 primary studies require as input previous *arguments and dialogues* from which infer information from. Two primary studies

Table 6 Studies grouped by input

Inputs	Studies #	Primary studies
Reserved, preferred and initial price values	32	PS1, PS2, PS3, PS8, PS9, PS10, PS12, PS13, PS14, PS15, PS16, PS17, PS18, PS20, PS25, PS26, PS27, PS28, PS29, PS30, PS31, PS34, PS35, PS37, PS38, PS39, PS41, PS42, PS45, PS46, PS47, PS48
Opponents' offers history	15	PS4, PS6, PS7, PS11, PS14, PS18, PS19, PS21, PS23, PS33, PS40, PS43, PS44, PS49, PS50
User's offers history	6	PS24, PS27, PS32, PS34, PS36, PS38
Arguments and dialogues	6	PS1, PS3, PS5, PS20, PS22, PS30
Rules	2	PS9, PS29

rely on input rules for the negotiation process, which the agents follow to generate offers.

Example – In PS2, the agents negotiate for purchasing or selling a laptop. The lowest price for buying the laptop and the highest price for selling it are already fixed before starting the negotiation as reserved values. The agents negotiate by exchanging price offers starting from an initial price value, e.g., “*I want to buy it for 50 points.*”. If the buyer agent bids higher than the reserved value of the seller agent, then the deal is automatically accepted.

Techniques used for automation— We present all techniques used in our selected primary studies to automate the negotiation process in Table 7. The total of the techniques mentioned in the table is less than the total number of primary studies, as some studies employ the same technique. However, some studies also employ more than one technique to automate the negotiation process.

Example – In PS4, two autonomous agents generate offers one by one to carry on the negotiation. The agents use deep reinforcement learning to learn the opponent's strategies and then utilize Bayesian learning to select one offer from the available offers to the agent. The study focuses on opponent modeling to learn about the opponent and generate counter-offers to end the negotiation by reaching an agreement or disagreement.

Type of agent involved in negotiation— For this parameter, we looked for the studies which suggest the approaches implemented on autonomous agents or at least involve one autonomous agent. Table 8 shows the division of our primary studies into two categories (1) studies involving only autonomous agents, and (2) studies involving one human agent and another autonomous agent.

Example – In PS1, the autonomous agent negotiates with other autonomous agents to form a team and complete the task without any human input as a cooperative game problem. The agents jointly perform the task and divide the final reward by communicating autonomously through a language channel using strings.

Table 7 Primary studies by used technique

Technique	Studies #	Primary studies
Reinforcement Learning (Q-Learning and DRL)	13	PS1, PS2, PS3, PS4, PS5, PS6, PS8, PS10, PS11, PS15, PS18, PS19, PS23
Optimization Approach	7	PS16, PS36, PS38, PS41, PS42, PS46, PS33
Gaussian Probability	5	PS7, PS21, PS24, PS32, PS40
Bayesian Learning	5	PS4, PS14, PS21, PS44, PS50
Neural Networks	4	PS14, PS37, PS43, PS49
Fuzzy Logic Systems	4	PS29, PS34, PS45, PS47
Monte Carlo Tree Search	3	PS9, PS21, PS48
Long Short-Term Memory Networks (LSTM)	2	PS1, PS30
Linear Programming	2	PS27, PS28
Argumentation Framework	2	PS20, PS22
Linear Regression	2	PS13, PS39
Heuristic Algorithm	2	PS26, PS31
Equilibrium Strategies	2	PS25, PS35
Genetic Algorithm	1	PS12
Markov Decision Process	1	PS24
Non-Linear Regression	1	PS14
Angle based Similarity Approach	1	PS17
Logistic Regression	1	PS37

However, in PS35, two human agents negotiate with an autonomous agent over the price of products. The human agents serve as customers while the autonomous agent serves as a seller. Each agent negotiates to maximize their reward in exchange for coins to purchase or sell the products and reach an agreement.

4.3 Limitation and future study (RQ3)

In the following, we discuss results for parameters related to the limitations found in the state of the art and future research directions.

Limitations and future research— The focus of the studies presented in the literature is based on collecting rewards where only two agents negotiate with each other. A total of 25 studies follow a bilateral negotiation strategy where one agent can only negotiate with one other agent.

Example – The primary study PS38 proposes a technique that can be applied to only two autonomous agents that negotiate for price where one agent represents a buyer and another agent represents a seller. This limitation can be addressed by proposing techniques for multilateral negotiation where multiple agents can negotiate with multiple opponents such as multiple buyers negotiating with the same seller or vice versa.

Another limitation in the state of the art is the improvement of the proposed technique. A total of 20 studies have suggested that the effectiveness of the proposed approach can be enhanced by combining it with another technique.

Example – The primary study PS8 uses Q-learning where two agents negotiate the price of electricity. The study suggests using inverse reinforcement learning for the same problem to better imitate human behavior in autonomous negotiation systems. Similarly, other studies consider changes in the proposed approaches.

Moreover, 19 studies consider the negotiation for a single issue. The studies propose techniques that only focus on negotiation for a single issue, e.g., negotiation only for the price as given in the same example above. This limitation can be addressed by proposing techniques for multiple real-life issues to develop intelligent systems that can be deployed in a real-world environment. For our mapping study, we consider studies that involve autonomous agents, hence there are still 12 studies that limit involving one human agent along with another autonomous agent in the negotiation process as mentioned in Table 8. This can be further addressed by proposing techniques that can be deployed on autonomous agents for automated negotiation.

In Table 9, we present the limitations found in the literature. It is noticed from the proposed research that the main idea behind the studies in the state of the art is focused on reward maximization. It will be interesting to propose techniques for automated negotiation where the agents negotiate real-life issues. For instance, the negotiation between two autonomous vehicles for one available parking lot, the negotiation between intelligent agents for automated auctions, and the negotiation between intelligent agents in healthcare. Another additional example can be making an ethical decision that involves no breach of privacy of passengers. Although nowadays the research community is focused on embedding ethics into automated decision making (Lo Piano 2020; Yu et al. 2018), we believe that no research is available that in real immerses the concept of ethics into automated decision making, which opens the doors for future research. We further discuss the need for an ethical autonomous system in Sect. 6.

5 Orthogonal findings

In this section, we discuss the results of data synthesis through horizontal analysis. As mentioned in Sect. 3.4, we utilized contingency tables to extract inter-parameter relationships among pairs of parameters in our classification framework. We computed and analyzed contingency tables for all pairs of parameters. However, for reasons of space, below we report only on the most relevant pairs. The total number of studies for inter-parameter relation mapping as a result of the contingency tables can be greater than the actual number of studies, i.e., 53 (including 50 primary studies together with three duplicate studies, that were merged in the final stage of the search and selection process, as shown in Fig. 1), as some studies focus on more than one parameter and are counted separately with the mapping of specific parameters.

Application domain—Negotiation purpose— Economics is the most focused application domain in the literature as shown in Fig. 2. The majority of studies that

Table 8 Studies grouped by agent type

Agent type	Studies #	Primary studies
Autonomous agents	38	PS1, PS2, PS3, PS4, PS5, PS6, PS7, PS8, PS9, PS11, PS14, PS15, PS16, PS17, PS18, PS19, PS21, PS22, PS23, PS25, PS27, PS30, PS31, PS33, PS34, PS37, PS38, PS39, PS40, PS41, PS42, PS43, PS44, PS45, PS46, PS48, PS49, PS50
Human and autonomous agents	12	PS10, PS12, PS13, PS20, PS24, PS26, PS28, PS29, PS32, PS35, PS36, PS47

propose techniques for reward maximization consider the application domain of the studies as economics. A total of 28 studies that focus on reward maximization for negotiation purposes consider economics as the application domain. Generic is the second most focused domain, with 9 studies that perform reward maximization for this purpose. The remaining application domains, such as cooperative environment, competing environment, and privacy, are focused on by only a small set of studies. This emphasis on economics reveals the need for a general-purpose agent, able to negotiate in multiple domains and for multiple purposes, as to enable the adoption of automated negotiation in the real world. Indeed, real-world scenarios are complex and involve negotiating for varied resources that cannot always be reduced to abstract economic scenarios (Lin et al. 2010).

Limitation—Negotiation purpose— As shown in Fig. 3, bilateral, single-issue negotiations and improving the proposed technique are the most observed limitations found in the selected primary studies. Indeed, the majority of the studies that focus on reward maximization present the limitation that the agents negotiate with only one opponent agent and only for a single kind of issue. A total of 20 studies that perform reward maximization suffer from negotiating for a single issue as a limitation, whereas another 20 studies that focus on reward maximization declare bilateral negotiation as an issue. Moreover, a total of 17 studies that focus on reward maximization declare the limitation of improving the techniques used for better performance. The studies that focus specifically on preference elicitation (3 studies) to represent the human user during negotiation suffer from the limitation that the user still has to be involved in the negotiation process, which needs to be resolved to achieve fully automated negotiation approaches. Moreover, even if the limitation of the single opponent is common in all studies that perform negotiation for multiple purposes, the single issue limitation is found only in 2 studies that focus on opponent modeling. This is due to the fact that these studies rely on the history of offers generated by an opponent agent to generate counteroffers. Therefore, the more issues are negotiated, the better the agent will be at generating strategies by learning the offers of the opponent.

Technique used—Negotiation purpose— Reinforcement learning is the most utilized approach in the literature by studies that perform negotiation for multiple purposes. The agents use reinforcement learning to learn opponent strategies and then leverage the acquired knowledge to generate offers strategically during negotiation attempting to secure maximum rewards for themselves. Moreover, as

reward maximization is the most focused negotiation purpose observed in the literature, reinforcement learning is the most utilized approach in the studies that perform negotiation to maximize rewards. As shown in Fig. 4, a total of 8 studies use this approach that perform negotiation for reward maximization, 7 studies focusing on opponent modeling and 4 studies that focus on cooperation during negotiation. The other most utilized approach in the literature is neural networks, which is utilized by 5 studies focusing on negotiation for reward maximization and 4 studies focusing on opponent modeling. The remaining techniques mentioned in the literature are utilized by studies with a small count as shown in Table 7.

Technique used—Application domain— As economics is the most observed application domain in the literature, it is the purpose for which most techniques are used. In the economics domain, neural networks are the most utilized technique, with 5 studies. It is followed by the optimization approach (4), reinforcement learning (3), and fuzzy logic (3). Most of the other techniques mentioned in Table 7 are used by a single study for economics. The other most observed application domain is generic, which utilizes reinforcement learning for 5, Bayesian learning for 4 and Gaussian probability for 3 studies. Fig. 5 presents the mapping of techniques used with the application domain as a result of our horizontal analysis.

Negotiation inputs—Negotiation purpose— The agents provide reserved and preferred values to initialize the negotiation process as mentioned in Sect. 4.2. We count them collectively here as offers with values for reward, which are most utilized by the studies that perform negotiation for reward maximization. As shown in Fig. 6, a total of 27 studies use offers for reward as input for the negotiation process for reward maximization. The offers generated by the opponent agents are also used as input by most of the studies to learn and generate counter offers. A total of 18 studies use opponent offers as input for opponent modeling, and 11 studies use these offers for reward maximization. For preference elicitation, 6 studies utilize the history of the user as input for the negotiation process. In 3 studies, when the agents negotiate for cooperation, additional dialogues are used as input along with offers for better explanation. In 2 studies, a set of rules is used as input for the negotiation to propose offers for reward maximization.

6 Discussion and future research challenges

In previous sections, we discussed the overview of the state of the art. In this section, we discuss research challenges that we concluded from the state of the art as our understanding for future research directions.

Is there more than economics for automated negotiation? Referring to Table 5, it is evident that most of the studies that propose automated negotiation techniques lie in the economics domain, in which agents negotiate for the price of an item being exchanged or a service being provided. One of the possible reasons why economics is the most observed domain is the encoding of the data that an agent needs to generate negotiation offers. Indeed, when negotiating for a price, the agents only need to encode the preferred and reserved price values that the agent wants to bid

Table 9 Studies grouped by limitation

Limitation	Studies #	Primary studies
Bilateral negotiation	25	PS2, PS4, PS6, PS7, PS8, PS10, PS11, PS16, PS17, PS18, PS19, PS25, PS27, PS31, PS33, PS34, PS37, PS38, PS44, PS45, PS46, PS47, PS48, PS49, PS50
Improve proposed methods	20	PS1, PS3, PS5, PS8, PS9, PS15, PS16, PS21, PS23, PS30, PS32, PS34, PS35, PS36, PS38, PS39, PS41, PS42, PS43, PS46
Single issue negotiation	19	PS8, PS12, PS16, PS17, PS20, PS22, PS26, PS27, PS28, PS31, PS34, PS35, PS38, PS39, PS41, PS42, PS46, PS47, PS49
Involves human agent	12	PS10, PS12, PS13, PS20, PS24, PS26, PS28, PS29, PS32, PS35, PS36, PS47
Time dependent negotiation	6	PS6, PS14, PS18, PS40, PS42, PS46

on and receive to maximize its reward. However, negotiating for resources of other kinds would require encoding different types of data, such as dialogues (Chen et al. 2022), rules (Malaby and Licato 2021), texts, images, or even data from sensors. One example is the work presented in PS36, in which the autonomous agent grants access to user data on her smartphone to an online service in exchange for a monetary reward. The agent negotiates the types of data to share, learning the user privacy preferences from the history of the previous interaction with other services. Hence, considering only prices reduces complexity during the negotiation process. We hypothesize that this is the reason why the economics setting is the most frequently considered by studies in the literature. However, to make such agents useful in many real life scenarios, it is important to develop intelligent autonomous agents able to negotiate in diverse domains, leveraging heterogeneous kinds of information.

Is it possible to have a general-purpose automated negotiating agent? As presented in Table 5, the majority of the studies propose techniques that focus on negotiation for domain-specific applications; This highlights an existing research challenge, identifiable in the need for a general-purpose negotiating agent that can generalize its behavior and adapt to multiple environments and resource types. An initial effort towards a general-purpose agent is the work by Lin and colleagues (Lin et al. 2010, 2014), in which the authors provide a framework to assist in the design of general-purpose negotiating agents, i.e., agents that “*can self-adapt in multiple environments by negotiating with other agents on real-life issues*”. While innovative and thought-provoking, the proposed framework hardly fills the gap in the literature

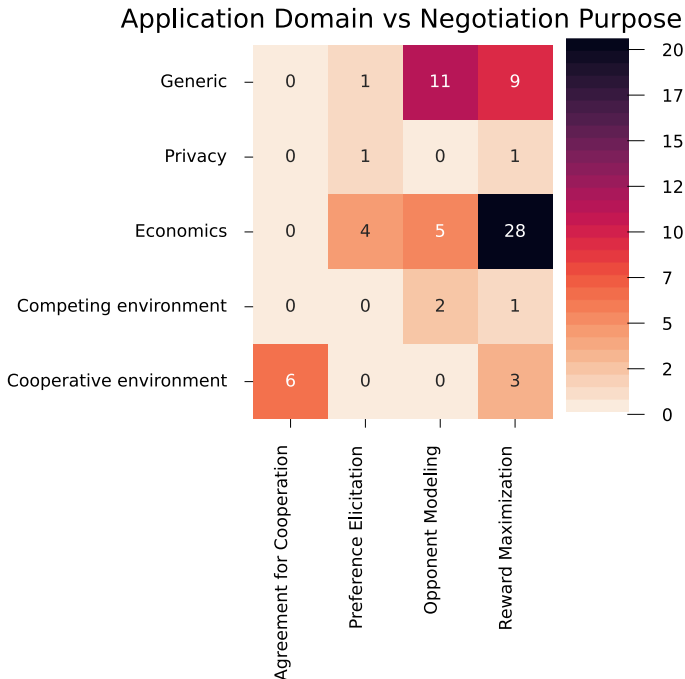


Fig. 2 Inter-parameter relationship between application domain and negotiation purpose



Fig. 3 Inter-parameter relationship between limitation and negotiation purpose

as agents built with the proposed framework exhibit severe limitations, such as the inability to negotiate with human negotiators. Hence, the challenge of a domain-independent autonomous agent capable of generalizing its behavior to negotiate in diverse environments is still relevant.

Is it possible to fully automate the negotiation process? As displayed in Table 8, there is a considerable number of studies that consider the negotiation between an autonomous agent and an human agent (Haim et al. 2017). In addition, in some studies, the human agent controls the decision of the autonomous agent, being able to change the decisions that the autonomous agent takes on her behalf (Baarslag et al. 2017). To fully automate the negotiation process where the agents will have full control on decisions without any human intervention, it is necessary to develop trustworthy autonomous agents that replicate human behavior and make ethical decisions even when things go wrong (Yazdanpanah et al. 2021). Indeed, human trust involves expectations of equality, justice, and fairness in the decision-making process (O’Fallon and Butterfield 2013), and the evaluation of such intelligent autonomous systems needs to minimize the risks that are posed to society when

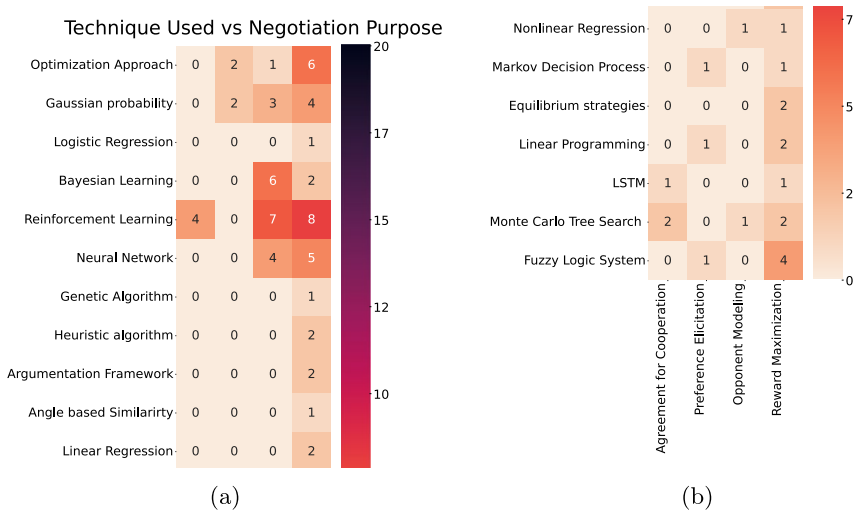


Fig. 4 Inter-parameter relationship between techniques used and negotiation purpose

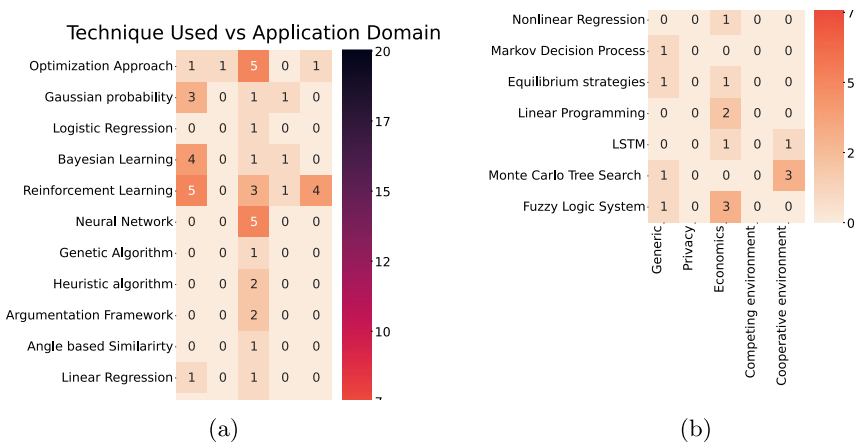


Fig. 5 Inter-parameter relationship between techniques used and application domain

biases are introduced in the automated decision-making process (Tolmeijer et al. 2020). More guidance is provided by the Ethics Guidelines for Trustworthy AI of the EU High-Level Expert Group on AI (High-Level Expert Group on AI 2019) that described the requirements that an AI system should satisfy to be considered ethical, among which: (1) respecting the rule of law; (2) being aligned with agreed ethical principles and values, including privacy, fairness, and human dignity; (2) ultimately keeping the humans in control, and (4) being robust and safe so that system's behavior remains trustworthy even if things go wrong. While these recommendations

are aimed at more general purpose AI systems, we believe that they are relevant for automated negotiation systems, especially considering the increasing importance that automated negotiation will acquire in a world of interacting autonomous systems. Hence, an open research direction is on how to introduce these principles in automated negotiation systems, to ultimately remove humans from the loop.

What are the growing research trends in the literature? A multitude of researchers are investigating automated negotiation techniques with different degrees of independence and different methodologies. We characterized the scientific interest of the research community on automated negotiation in terms of relevant venues. As mentioned in Sect. 3.2, we have selected top-level venues for conference proceedings and journal publications for this mapping study. Looking at Table B, we can observe that the most popular venues for works in the domain of automated negotiation are AI-focused ones. If, on the one hand, this can be motivated by the growing interest that AI technologies are receiving in recent years, on the other hand it highlights that automated negotiation could play a central role in future AI-based autonomous systems. Indeed, future autonomous systems will be able to represent humans and make decisions on their behalf without their intervention and will interact with other humans or autonomous systems (Matthews et al. 2021). Hence, in the future, these systems could take advantage of negotiation techniques to achieve more equitable and fair interactions with humans and other systems.

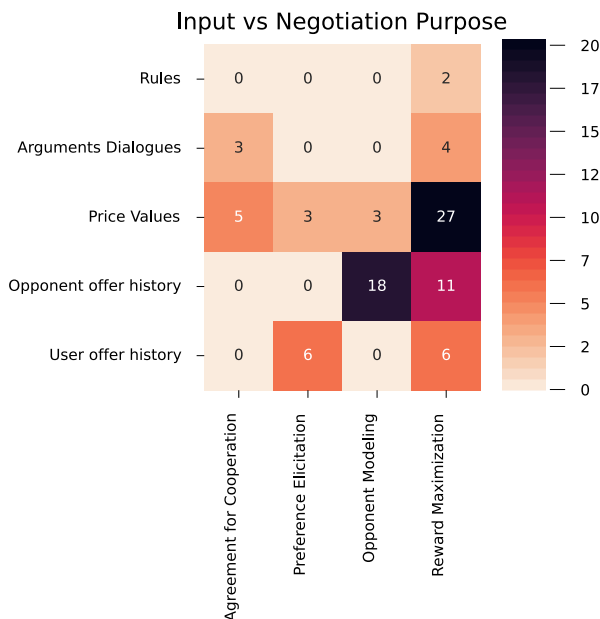


Fig. 6 Inter-parameter relationship between negotiation inputs and negotiation purpose

7 Threats to validity

In this section, we discuss the quality of the data and threats to the validation of primary studies. At first, we defined a research protocol to verify the high quality of the acquired data. The research protocol was established before proceeding with the data collection and was designed by following the guidelines for planning the systematic mapping study from Keele et al. (2007). As presented in Sect. 3, we rigidly complied with the research protocol during the evolution of the proposed study. In addition, we combined and executed all steps involved in the research protocol (e.g., study design, search and selection, data extraction, and data analysis) together in teams to additionally guarantee the quality attributes of the established research protocol. In the remainder of this section, we further discuss threats to validity in the context of our study and how we did our best to diminish them.

External validity. This stage deals with validating the risk of quality assurance (Wohlin et al. 2012), i.e., whether the proposed study correctly identified the state-of-the-art literature or not. To mitigate this type of risk, we selected top-level venues for searching publications where we manually applied keyword-based search and selection processes for conference proceedings and journal publications. One possible threat to the external validity of our study is the possibility that, despite our efforts, some relevant venues were not included in the ranking used as a base for this selection process. However, this threat is mitigated by the subsequent snowballing process, during which forward and backward snowballing have been applied to extend the set of primary studies in order to have a comprehensive set of studies representative of the literature of the field. Furthermore, we only extracted peer-reviewed publications and eliminated short papers, preprints, and editorials. This process was applied to maintain quality, as peer-reviewed studies are an essential item for high-grade publications. Subsequently, we applied a set of inclusion and exclusion criteria, which led to further refinement of the proposed study.

Internal validity. This stage deals with validating the risk related to the study design (Wohlin et al. 2012). To mitigate this type of risk, we established a research protocol and a classification framework. The classification framework is designed and strictly followed as detailed in Sect. 3, where the design of our study is presented from a methodological point of view. The researchers involved in the proposed study have independently followed the framework to adopt fairly evaluated descriptive statistics of the research. Subjective biases and interpretations were mitigated by closely complying with the selection criteria to evaluate the studies. In addition, meetings were held regularly during the selection and analysis process to jointly discuss examples and doubts. Regarding the validity of the synthesis of the collected data, well-assessed descriptive statistics have been employed for both the vertical and horizontal analysis, so the threats are minimal. Furthermore, during the horizontal analysis, we cross-analyzed the values of different concepts and attributes of the classification framework to perform a sanity test of the extracted data. This analysis confirmed the consistency of the extracted data.

Construct validity. This stage focuses on ensuring the justification and compatibility of the primary studies with the research questions (Wohlin et al. 2012). In

the context of mapping studies, it mainly relates to ensuring that the selected primary studies were suited to answer the research questions. To ensure that, we manually extracted the literature from top-level venues related to computer science and applications, without relying on a single scientific search engine. This allowed us to avoid potential biases due to publishers' policies and business concerns. During this step, a keyword-based search was performed. We are reasonably confident about the construction of the search query since the terms used are very general and suited to identify studies related to our research questions. The results of this process were further expanded by applying forward and backward snowballing. Subsequently, a screening of identified studies was performed in accordance with the inclusion and exclusion criteria. Furthermore, as suggested by Wohlin (Wohlin et al. 2012), a random set of 10 studies was extracted and analyzed by all authors in order to guarantee alignment of the analyses.

Conclusion validity. In this stage, the focus is to ensure the validity of the conclusion derived based on primary studies (Wohlin et al. 2012). To do that, we mitigated potential sources of bias during the data extraction and analysis processes by strictly adhering to our previously established research protocol, specifically designed to extract data useful to answer our research questions. Furthermore, to reduce biases, we followed well-established guidelines for systematic mapping studies (Kitchenham and Brereton 2013; Petersen et al. 2008a, 2015) to diminish the risk regarding the validity of the conclusion. Finally, collected data and employed scripts are made available in the replication package to ensure the transparency of our research.

8 Conclusion

The proposed systematic mapping study (SMS) analyzes relevant studies on automated negotiation for autonomous systems. For this purpose, we extracted 50 primary studies from 73,760 relevant studies through a selection process from the top-ranked publication venues. We applied keyword-based search to extract relevant studies from 24 conferences and 22 journals, followed by backward and forward snowballing. We follow the precise inclusion and exclusion criterion to include only studies that focus on the practical implementation of the techniques proposed for automated negotiation.

We designed a classification framework to identify, in particular, (1) the specific purpose of the negotiation, (2) the required inputs and expected outputs of the negotiation process together with the techniques applied to automate the process, (3) the limitation of the state of the art and the future research directions, and (4) possible directions for consolidating the existing literature on negotiation with a focus on automation.

We present the key results of our systematic mapping study by introducing a blend of content analysis and narrative synthesis (vertical analysis), and correspondence analysis through contingency tables (horizontal analysis). We found that 27

studies in the literature primarily focus on economics, raising questions about potential applications in other domains; 12 studies involve human agents, raising concerns about fully automating the negotiation. This emphasizes the importance of building trust in these systems, as autonomous systems are designed to make decisions on the user's behalf. We have also discussed the significance of these aspects for future research. The growing interest in AI technologies highlights that automated negotiation can play a central role in future AI-based autonomous systems. Indeed, future autonomous systems will act on humans' behalf and make decisions for them in a fully automated way while interacting with other humans or other autonomous systems (Matthews et al. 2021). Hence, in the future, these systems can benefit from negotiation techniques to automatically achieve more equitable and fair interactions with humans and other systems.

This study further helps the research community find potential studies published in the domain of automated negotiation that also consider practical implementation aspects. To the best of our knowledge, no research such as our systematic mapping study is available that introduces a set of potential studies regarding automated negotiation that provides the practical implementation details of the proposed techniques. Hence, we believe that our study establishes a valuable asset for both practitioners and researchers in the wide spectrum of autonomous systems that involve intelligent agents aiming to reach an agreement to automatically find a solution that satisfies all involved parties.

Appendix A. Primary studies

See Table 10.

Table 10 Primary studies

ID	Title	Author	Year
PS1	Coalitional Negotiation Games with Emergent Communication (Gao et al. 2022)	Xiaoyang Gao, Siqi Chen, Lin Jie, Yang Yang, Haiying Wu, and Jianye Hao	2022
PS2	Deep Learnable Strategy Templates for Multi-Issue Bilateral Negotiation (Bagga et al. 2022)	Pallavi Bagga, Nicola Paoletti, and Kostas Stathis	2022
PS3	Deep reinforcement learning with emergent communication for coalitional negotiation games (Chen et al. 2022)	Siqi Chen, Yang Yang, and Ran Su	2022
PS4	A Bayesian Policy Reuse Approach for Bilateral Negotiation Games (Gao et al. 2022)	Xiaoyang Gao, Siqi Chen, Qisong Sun, Yan Zheng, and Jianye Hao	2022
PS5	Distributed Emergent Agreements with Deep Reinforcement Learning (Schmidt et al. 2021)	Kyrill Schmid, Robert Müller, Lenz Belzner, Johannes Tochtermann, and Claudia Linhoff-Popien	2021
PS6	An Autonomous Negotiating Agent Framework with Reinforcement Learning based Strategies and Adaptive Strategy Switching Mechanism (Sengupta et al. 2021)	Ayan Sengupta, Yasser Mohammad, and Shinji Nakadai	2021
PS7	Convergence of probabilistic automatic negotiation: mutual maximum likelihood estimation (Tsumura 2021)	Koji Tsumura	2021
PS8	A context-aware approach to automated negotiation using reinforcement learning (Kröhlhng et al. 2021)	Dan E Kröhlhng, Omar JA Chiotti, and Ernesto C Martínez	2021
PS9	Exploring Monte Carlo Negotiation Search with Nontrivial Agreements (Malaby and Licato 2021)	Elijah Alden Malaby and John Licato	2021
PS10	A deep reinforcement learning-based agent for negotiation with multiple communication channels (Gao et al. 2021)	Xiaoyang Gao, Siqi Chen, Yan Zheng, and Jianye Hao	2021
PS11	Detecting and Learning Against Unknown Opponents for Automated Negotiations (Wu et al. 2021)	Leling Wu, Siqi Chen, Xiaoyang Gao, Yan Zheng, and Jianye Hao	2021
PS12	A Bilevel Game Model for Ascertaining Competitive Target Prices for a Buyer in Negotiation with Multiple Suppliers (Kumar et al. 2022)	Akhilesh Kumar, Anjana Gupta, and Aparna Mehra	2021
PS13	A Supervised Topic Model Approach to Learning Effective Styles within Human-Agent Negotiation (Xu et al. 2020)	Yuyu Xu, David Jeong, Pedro Sequeira, Jonathan Gratch, Javed Aslam, and Stacy Marsella	2020
PS14	Modeling Opponent Strategy in Multi-Issue Bilateral Automated Negotiation Using Machine Learning (Mohammadi et al. 2020)	Fatemeh Mohammadi Ashmani, Zahra Movahedi, and Kazim Fouladi	2020

Table 10 (continued)

ID	Title	Author	Year
PS15	Negotiating team formation using deep reinforcement learning (Bachrach et al. 2020)	Yoram Bachrach, Richard Everett, Edward Hughes, Angeliki Lazaridou, Joel Z Leibo, Marc Lanctot, Michael Johanson, Wojciech M Czarnecki, and Thore Graepel	2020
PS16	A hybrid concession mechanism for negotiating software agents in competitive environments (Mansour 2020)	Khalid Mansour	2020
PS17	Agent-based cloud service negotiation architecture using similarity grouping approach (Rajavel et al. 2020)	Rajkumar Rajavel, Sathish Kumar Ravichandran, and GR Kanagachidambaresan	2020
PS18	Deep reinforcement learning for acceptance strategy in bilateral negotiations (Razeghi et al. 2020)	Yousef Razeghi, Celal Ozan Berk Yavuz, and Reyhan Aydoğan	2020
PS19	RLBOA: A Modular Reinforcement Learning Framework for Autonomous Negotiating Agents (Bakker et al. 2019)	Jasper Bakker, Aron Hammond, Daan Bloembergen, and Tim Baarslag	2019
PS20	Argumentation-based Negotiation with Incomplete Opponent Profiles (Dimopoulos et al. 2019)	Yannis Dimopoulos, Jean-Guy Mailly, and Pavlos Moraitis	2019
PS21	MCTS-based Automated Negotiation Agent (Buron et al. 2019)	Cédric LR Buron, Zahia Guessoum, and Sylvain Ductor	2019
PS22	Numerical Abstract Persuasion Argumentation for Expressing Concurrent Multi-Agent Negotiations (Arisaka and Ito 2019)	Ryuta Arisaka and Takayuki Ito	2019
PS23	Meta-Strategy for Multi-Time Negotiation: A Multi-Armed Bandit Approach (Kawata and Fujita 2019)	Ryohei Kawata and Katsuhide Fujita	2019
PS24	Automated Negotiation with Gaussian Process-based Utility Models (Leahu et al. 2019)	Haralambie Leahu, Michael Kaisers, and Tim Baarslag	2019
PS25	Negotiation Strategies for Agents with Ordinal Preferences (Erllich et al. 2018)	Sefi Erlich, Noam Hazon, and Sarit Kraus	2018
PS26	One-to-Many Multi-agent Negotiation and Coordination Mechanisms to Manage User Satisfaction (Najjar et al. 2018)	Anro Najjar, Yazan Mualla, Kamal Singh, and Gauthier Picard	2018
PS27	Automated Negotiations Under User Preference Uncertainty: A Linear Programming Approach (Tsimpoukis et al. 2018)	Dimitrios Tsimpoukis, Tim Baarslag, Michael Kaisers, and Nikolaos G Paterakis	2018
PS28	A systematic model of stable multilateral automated negotiation in e-market environment (Gao et al. 2018)	Taiguang Gao, Min Huang, Qing Wang, Mingqiang Yin, Wai Ki Ching, Loo Hay Lee, and Xingwei Wang	2018

Table 10 (continued)

ID	Title	Author	Year
PS29	A multi-demand negotiation model based on fuzzy rules elicited via psychological experiments (Zhan et al. 2018)	Jieyu Zhan, Xudong Luo, Cong Feng, and Minghua He	2018
PS30	Emergent Communication through Negotiation (Cao et al. 2018)	Kris Cao, Angeliki Lazaridou, Marc Lanctot, Joel Z Leibo, Karl Tuyls, and Stephen Clark	2018
PS31	Concurrent bilateral negotiation for open e-markets: the CONAN strategy (Alrayes et al. 2018)	Bedour Alrayes, Özgür Kafalı, and Kostas Stathis	2018
PS32	The Value of Information in Automated Negotiation: A Decision Model for Eliciting User Preferences (Baarslag and Kaisers 2017)	Tim Baarslag and Michael Kaisers	2017
PS33	POPPONENT: Highly accurate, individually and socially efficient opponent preference model in bilateral multi issue negotiations (Zafari and Nassiri-Mofakham 2016)	Farhad Zafari and Faria Nassiri-Mofakham	2017
PS34	Designing an intelligent decision support system for effective negotiation pricing: A systematic and learning approach (Fu et al. 2017)	Xin Fu, Xiao-Jun Zeng, Xin Robert Luo, Di Wang, Di Xu, and Qing-Liang Fan	2017
PS35	Human-computer negotiation in a three player market setting (Haim et al. 2017)	Galit Haim, Bo An, Sarit Kraus, et al	2017
PS36	An Automated Negotiation Agent for Permission Management (Baarslag et al. 2017)	Tim Baarslag, Alan Alper, Richard Gomer, Muddasser Alam, Perera Charith, Enrico Gerding, et al	2017
PS37	Algorithm selection in bilateral negotiation (Ilany and Gal 2016)	Litan Ilany and Ya'akov Gal	2016
PS38	An Agent Architecture for Concurrent Bilateral Negotiations (Alrayes and Stathis 2013)	Bedour Alrayes and Kostas Stathis	2013
PS39	Complex and Concurrent Negotiations for Multiple Interrelated e-Markets (Sim 2012)	Kwang Mong Sim	2012
PS40	Using Gaussian Processes to Optimise Concession in Complex Negotiations against Unknown Opponents (Williams et al. 2011)	Colin Richard Williams, Valentin Robu, Enrico Harm Gerding, and Nicholas Robert Jennings	2011
PS41	Automated Negotiation with Decommittment for Dynamic Resource Allocation in Cloud Computing (An et al. 2010)	Bo An, Victor Lesser, David Irwin, and Michael Zink	2010
PS42	A Multilateral Negotiation Model for Cloud Service Market (Yoo and Sim 2010)	Dongjin Yoo and Kwang Mong Sim	2010
PS43	An Opponent's Negotiation Behavior Model to Facilitate Buyer-seller Negotiations in Supply Chain Management (Fang et al. 2008)	Fang Fang, Ye Xin, Xia Yun, and Xu Haitao	2008

Table 10 (continued)

ID	Title	Author	Year
PS44	Opponent Modelling in Automated Multi-Issue Negotiation Using Bayesian Learning (Hindriks and Tykhonov 2008)	Koen Hindriks and Dmytro Tykhonov	2008
PS45	A fuzzy constraint based model for bilateral, multi-issue negotiations in semi-competitive environments (Luo et al. 2003)	Xudong Luo, Nicholas R Jennings, Nigel Shadbolt, Hofung Leung, and Jimmy Ho-man Lee	2003
PS46	Optimal Negotiation Strategies for Agents with Incomplete Information (Fatima et al. 2001)	S Shaheen Fatima, Michael Wooldridge, and Nicholas R Jennings	2001
PS47	On Fuzzy E-Negotiation Agents: Autonomous Negotiation with Incomplete and Imprecise Information (Kowalczyk and Bui 2000)	Ryszard Kowalczyk and Van Bui	2000
PS48 (a)	Strategic Negotiations for Extensive-Form Games (De Jonge and Zhang 2020)	Dave De Jonge and Dongmo Zhang	2020
PS48 (b)	Automated Negotiations for General Game Playing (De Jonge and Zhang 2017)	Dave De Jonge and Dongmo Zhang	2017
PS49 (a)	A Deep Reinforcement Learning Approach to Concurrent Bilateral Negotiation (Bagga et al. 2020)	Pallavi Bagga, Nicola Paoletti, Bedour Alrayes, and Kostas Stathis	2021
PS49 (b)	ANEGMA: an automated negotiation model for e-markets (Bagga et al. 2021)	Pallavi Bagga, Nicola Paoletti, Bedour Alrayes, and Kostas Stathis	2021
PS50 (a)	Decoupling Negotiating Agents to Explore the Space of Negotiation Strategies (Baarslag et al. 2014)	Tim Baarslag, Koen Hindriks, Mark Hendrikx, Alexander Dirkzwager, and Catholijn Jonker	2014
PS50 (b)	A Tit for Tat Negotiation Strategy for Real-Time Bilateral Negotiations (Baarslag et al. 2013)	Tim Baarslag, Koen Hindriks, and Catholijn Jonker	2013

Appendix B. List of selected venues for keyword-based search

See Table 11.

Table 11 Selected venues with total number of studies

Conference	No. of studies	Journals	No. of studies
Conference on Artificial Intelligence (AAAI)	7539	Foundations and Trends in Machine Learning	23
Adaptive Agents and Multi-Agents Systems (AAMAS)	2320	IEEE Transactions on Pattern Analysis and Machine Intelligence	1449
International Conference on Artificial Intelligence and Statistics (AISTATS)	1622	IEEE Computational Intelligence Magazine	268
International Joint Conference on Automated Reasoning (CADE)	106	Wiley Interdisciplinary Reviews: Data Mining and Knowledge Discovery	223
Conference on Human Factors in Computing Systems (CHI)	7408	IEEE Transactions on Neural Networks and Learning Systems	2267
Acm Conference on Economics and Computation (EC)	485	Nature Machine Intelligence	410
European Conference on Artificial Intelligence (ECAI)	1546	Artificial Intelligence Review	589
International Conference on Machine Learning (ICML)	4115	Computers in Human Behavior	2600
International Joint Conference on Artificial Intelligence (IJCAI)	4549	Computer Science Review	217
IEEE International Joint Conference on Neural Networks (IJCNN)	4477	International Journal of Intelligent Systems	845
Conference in Uncertainty in Artificial Intelligence (UAI)	660	Neural Networks	1514
IEEE Conference on Systems, Man and Cybernetics (SMC)	3323	Journal of Computer-Mediated Communication	125
ACM Symposium on Applied Computing (ASM SAC)	1511	IEEE Transactions on Emerging Topics in Computational Intelligence	338
International Conference on Advanced Information Systems Engineering (CAISE)	638	ACM Transactions on Intelligent Systems and Technology	350
Conference on Decision and Control (CDC)	4974	IEEE Transactions on Software Engineering	525
IEEE Congress on Evolutionary Computation (CEC)	1892	Decision Support Systems	647

Table 11 (continued)

Conference	No. of studies	Journals	No. of studies
ACM Conference on Computer Supported Cooperative Work (CSCW)	672	Engineering Applications of Artificial Intelligence	1550
Automated Software Engineering (ASE)	830	International Journal of Neural Systems	356
ACM SIGSOFT Conference on The Foundations of Software Engineering (ESEC/FSE)	896	Artificial Intelligence	511
Genetic and Evolutionary Computation Conference (GECCO)	2414	Robotics and Autonomous Systems	979
Human-Robot Interaction (HRI)	1467	Journal of Artificial Intelligence and Soft Computing Research	90
International Conference on Software Engineering (ICSE)	1948	Journal of Machine Learning Research	1065
IEEE International Symposium on Robot and Human Interactive Communication (RO-MAN)	1002		
Robotics: Science and systems (RSS)	425		
Total	56,819	Total	16,941

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Declarations

Conflict of interest The authors declare that they have no Conflict of interest.

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Authors and Affiliations

Mashal Afzal Memon¹ · Gian Luca Scoccia² · Marco Autili¹

✉ Mashal Afzal Memon
mashalafzal.memon@univaq.it

Gian Luca Scoccia
gianluca.scoccia@gssi.it

Marco Autili
marco.autili@univaq.it

¹ Department of Information Engineering, Computer Science, and Mathematics, University of L'Aquila, L'Aquila, Italy

² Department of Computer Science, Gran Sasso Science Institute, L'Aquila, Italy